AI-1 Spring 26 Project Proposal

Team Members

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Team ID (32)

Idea Short Name (Shawqi (shogi))

Idea Description

Shogi (Japanese Chess) is a complex board game that involves deep strategic planning and prediction of opponent moves.

This project focuses on building a fully interactive Shogi game integrated with an AI opponent that adapts its behavior based on selected difficulty levels.

Agent Architecture

☐ Simple Reflex Agent

☐ Model-Based Reflex Agent

√ Goal-Based Agent

√ Utility-Based Agent

Problem-Solving Framework

√ Classical Search

☐ Local Search (Optimization)

√ Adversarial Search (Two-Player Games)

☐ Constraint Satisfaction Problem

Proposed Algorithm(s)

Alpha-Beta

PEAS

Performance Measures	Winning the game (Checkmate) Maximizing board evaluation score Minimizing opponent advantage Fast response time
Environment	Shogi board (9x9) Pieces Human player AI opponent
Actuators	Making a move Capturing opponent pieces Promoting pieces
Sensors	Current board state Available moves Opponent moves

Environment Properties

√ Fully Observable ☐ Partially Observable	√ Deterministic ☐ Stochastic √ Strategic	√ Discrete ☐ Continuous
☐ Single-Agent √ Multi-Agent	√ Sequential ☐ Episodic	√ Static ☐ Dynamic